

# José Fabián Sifuentes Gálvez

Electronic Designer | Embedded Systems | RF & Satellite Communications

jf19sifuentesg@gmail.com · +51 953 186 335 · Lima, Peru · github.com/noonejf · noonejf.github.io

## Profile

Mechatronics Engineer (PUCP) specialized in electronic design and embedded systems, with 3+ years of experience in multi-layer PCBs, C/C++ firmware on STM32 and Raspberry Pi, and RF communications for satellite missions. Experienced across the full hardware cycle: schematic capture, routing, physical prototype validation, sensor and peripheral integration, and UHF link testing. Proven track record in space projects (CubeSats, ground stations) and commercial products.

## Experience

**Professional Intern — Space Engineering** · INRAS – PUCP · pre-professional until 2025 Jul 2024 – Present

- ▶ Developed UHF communication systems for satellite missions, including link testing.
- ▶ Designed and developed the INTISAT ground station interface with real-time telemetry.
- ▶ Built 3D simulations of space tethers and integrated data link protocols.

**Developer — Embedded Systems & Interfaces** · Bear Tech Jan – Jun 2025

- ▶ Internal pipeline-inspection robot: integration of 6 cameras (4 in a 360° array) on Raspberry Pi, IMU + odometry navigation (STM32, UART), and TCP/IP teleoperation.
- ▶ Automatic structural-fault detection with YOLOv8 integrated into the operator HUD (PyQt5).
- ▶ Designed the robot's electronic circuitry: component selection and integration of power and control stages.
- ▶ Designed multilayer PCBs with EasyEDA: schematics, routing, and physical prototype validation.

**Student Researcher — Bionics** · LIBRA – PUCP Feb – Sep 2024

- ▶ Technical analysis of upper-limb myoelectric prostheses (electronics and actuation).
- ▶ OCTAHAND project: prosthesis recovery, repair, and improvement.

**Fullstack Developer** · Freelance / Commercial Project 2022 – Present

- ▶ Project management web application with React/Node.js on AWS — full hardware-software perspective.

## Featured Projects

**AYNISAT — Peruvian CubeSat (CONIDA)** · INRAS 2024 – Present

- ▶ Communications payload with STM32: complete electronic design in EasyEDA and UHF link testing.

**S-Band Transceiver — RF Design** · INRAS 2025 – Present

- ▶ RF chain design: amplifier + FPGA, DAC filter, modulator, clock PLL, TCXO, and patch antenna.
- ▶ Link budget analysis and justified component selection for the system.

**INTISAT — UHF Communications Payload Lead** · APSCO competition 2024 – Present

- ▶ Telemetry system, satellite communications, and ground control station.

**Digital PID Servo Control System** · academic project, PUCP 2025

- ▶ PID controller design under voltage constraints: MATLAB/Simulink synthesis and embedded deployment (PlatformIO) validated with real trajectories.

## Technical Skills

**Electronic Design:** EasyEDA, KiCad · multilayer PCBs: schematics, routing, DRC, prototype validation

**Embedded:** STM32, Raspberry Pi 5, Arduino, FPGA (integration) · C/C++ firmware · PlatformIO

**Buses & Protocols:** I2C, SPI, UART, CAN · satellite telemetry, UHF links

**RF & Communications:** link budget, amplifiers, filters, PLL, TCXO, patch antennas, modulation

**Control & Simulation:** MATLAB/Simulink, PID design, ROS2, Gazebo, CoppeliaSim · robot kinematics

**Software:** Python, C, C++, MATLAB · PyQt5, Git, Linux/Windows

**Mechanical Design:** SolidWorks, Fusion 360, Autodesk Inventor, ANSYS

## Education, Languages & Certifications

**Mechatronics Engineering** · Pontificia Universidad Católica del Perú (PUCP) 2020 – 2025 · Lima, Peru

**Languages:** Spanish (native) · English — TOEFL (ICPNA)

**Certifications:** PERUMEC 2025 — robotics competitor · UNI Space TechWeek — speaker (LINKU project)